Module 4: The Transport Layer

4.1 Why a transport layer?

The network layer moves packets between hosts on a best-effort basis: packets may be lost, duplicated or reordered. The transport layer provides logical communication between processes. UDP adds only multiplexing and a checksum. TCP adds connection setup, reliable in-order delivery, flow control and congestion control.

4.2 Reliable data transfer

Reliability is built from sequence numbers, acknowledgements, timers and retransmission. Go-Back-N retransmits every unacknowledged segment after a timeout; Selective Repeat retransmits only the missing segment. TCP uses cumulative acknowledgements with a single retransmission timer, and fast retransmit after three duplicate ACKs.

The retransmission timeout is computed from a smoothed round-trip time estimate: EstimatedRTT = (1 - a) * EstimatedRTT + a * SampleRTT with a = 0.125, DevRTT = (1 - b) * DevRTT + b * |SampleRTT - EstimatedRTT| with b = 0.25, and TimeoutInterval = EstimatedRTT + 4 * DevRTT.

4.3 Flow control

Flow control stops a sender from overrunning the receiver's buffer. The receiver advertises a receive window rwnd in every segment; the sender keeps the amount of unacknowledged data below rwnd.

4.4 Congestion control

Congestion control stops senders from overrunning the network. TCP keeps a congestion window cwnd and sends at most min(cwnd, rwnd) unacknowledged bytes. In slow start, cwnd starts at 1 MSS and doubles every round-trip time until it reaches the slow-start threshold ssthresh. In congestion avoidance, cwnd grows by about 1 MSS per round-trip time (additive increase). On three duplicate ACKs, TCP Reno halves cwnd and enters fast recovery (multiplicative decrease). On a timeout, ssthresh is set to cwnd / 2 and cwnd returns to 1 MSS.

The resulting sawtooth is called AIMD: additive increase, multiplicative decrease. AIMD converges to a fair share when several flows compete for one bottleneck link. TCP CUBIC, the Linux default, replaces linear growth with a cubic function of the time since the last loss, which recovers bandwidth faster on long fat networks.

4.5 Throughput

Average TCP throughput is about 0.75 * W / RTT, where W is the window size in bytes when loss occurs. Under a loss rate L, throughput is approximately 1.22 * MSS / (RTT * sqrt(L)).